



WESTFIELDS SPORTS

YEAR 12 TRIAL EXAM

2008

MATHEMATICS

Reading time – 5 minutes

Working Time - 3 hours

INSTRUCTIONS TO STUDENTS:

- check that you have the correct paper
- approved calculators may be used
- use a new page for the start of each question
- all necessary working must be shown
- start each question on a new page

Total marks - 120

Attempt Questions 1 - 10

All questions are of equal value

Question 1**Marks**

a) Evaluate, correct to three significant figures,

2

$$\sqrt{\frac{(3.024)^3}{25.5 - 13.018}}$$

b) Solve for x : $x^3 = 4x^2$

2

c) Find the primitive of: $\frac{1}{3e^x}$

1

d) Simplify: $\frac{1}{m^2 - 4m + 3} - \frac{1}{m^2 - 1}$

3

e) Solve the pair of simultaneous equations

2

$$x - 2y = 1$$

$$xy = 1$$

f) Find the integers a and b such that $\frac{1}{2 - \sqrt{3}} = a + b\sqrt{3}$

2

Question 2

a) Write down the derivatives of:

(i) $(3x + 4)^7$

2

(ii) $x^3 e^x$

2

(iii) $\frac{3x}{\sin x}$

2

b) Find the exact value of $\tan \frac{\pi}{3} + \operatorname{cosec} \frac{\pi}{4}$

2

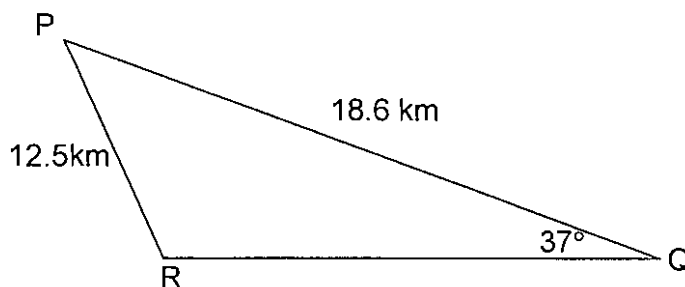
c) Consider the quadratic function $x^2 - (k + 2)x + 4 = 0$

2

For what value of k does the quadratic function have real roots?.

d)

2

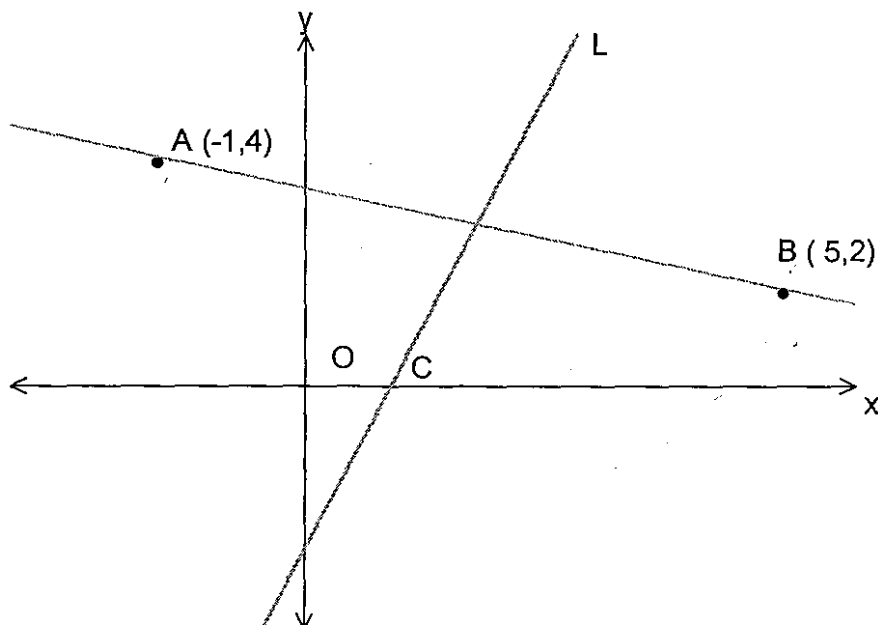


In the diagram above, $PQ = 18.6$ km, $PR = 12.5$ km and $\angle PQR = 37^\circ$. $\angle PRQ$ is obtuse. Find the size of $\angle PRQ$ correct to the nearest minute.

Question 3

Marks

a)



The diagram above shows the points $A(-1, 4)$ and $B(5, 2)$. The line L has equation $3x - y - 3 = 0$ and cuts the x -axis at C .

- (i) Show that the length of AB is $2\sqrt{10}$ units. 1
- (ii) Find the coordinates of M , the midpoint of AB . 1
- (iii) Find the gradient of AB . 1
- (iv) Show that the equation of AB is $x + 3y - 11 = 0$ 1
- (v) Prove that L is perpendicular bisector of AB . 2
- (vi) Find the coordinates of C . 1
- (vii) Write down the equation of the circle with AB as a diameter. 1

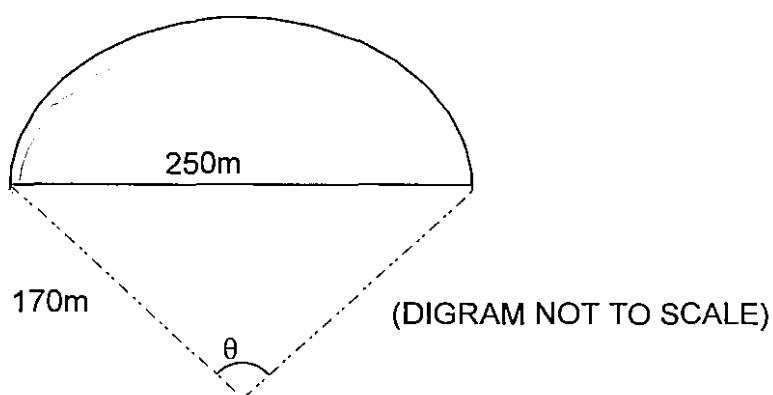
b) α and β are the roots of the equation $x^2 - 6x + 10 = 0$. Find the values of:

- (i) $\alpha + \beta$ 1
- (ii) $\alpha\beta$ 1
- (iii) $(\alpha + 1)(\beta + 1)$ 2

Question 4

Marks

a) A straight road was constructed to cut a dangerous bend on a country road. It was found that the bend was part of an arc of radius 170 metres and the straight road was 250 metres long.



- | | | |
|------|--|---|
| (i) | Use the cosine rule to find the size of θ correct to the nearest degree. | 2 |
| (ii) | Find the distance by which the old road was shortened.
Answer correct to the nearest metre. | 3 |
- b) For the parabola $16y = x^2$, write down the:
- | | | |
|------|----------------------------|---|
| (i) | coordinates of the focus | 2 |
| (ii) | equation of the directrix. | 1 |
- c) (i) Sketch the graph of $y = -2\cos x$ for $0 \leq x \leq 2\pi$
- | | | |
|------|---|---|
| (ii) | On the same axes, sketch the graph of $y = -2\cos x - 1$ for $0 \leq x \leq 2\pi$ | 2 |
|------|---|---|

Question 5

Marks

a) Find the equation of the

(i) tangent and

4

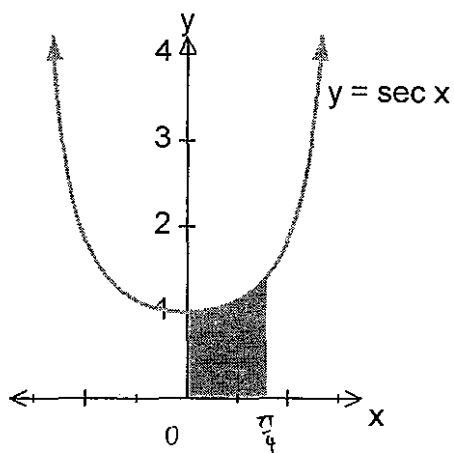
(ii) the normal to the curve $y = x \sin x$ at the point $(\frac{\pi}{2}, \frac{\pi}{2})$

2

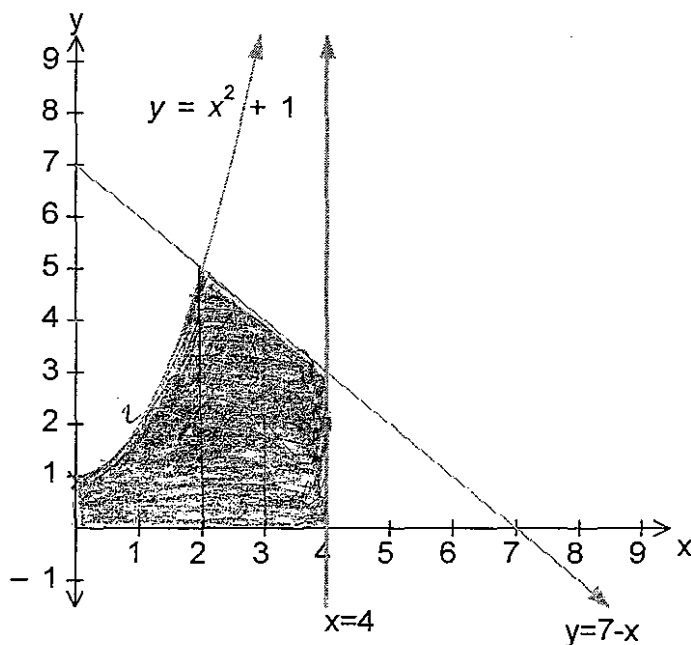
b) The shaded region which lies between the x axis and the curve $y = \sec x$

3

from $x = 0$ to $x = \frac{\pi}{4}$ is rotated about the x axis to form a solid. Find the volume of the solid.



c)



Use Simpson's Rule with five function values to find an approximation of the Shaded area.

3

Question 6**Marks**

a) Given that $\sin \theta = \frac{3}{4}$ and $0^\circ < \theta < 90^\circ$, find as a single expression with rational denominator, the exact value of:

(i) $\cos \theta$ 2

(ii) $\cos \theta + \tan \theta$ 2

b) Find all values of θ such that $\sin 2\theta = 1$ and $0 \leq \theta \leq 2\pi$ 2

c) Solve the inequality $4x - x^2 > 0$ 2

d) For what value of m does the line $y = m(x+1)$ have no intersection with the parabola $y = 2x^2$? 2

e) Solve the equation $e^x - 9e^{-x} = 0$ 2

Question 7.

a) The function $f(x)$ is defined by the rule $f(x) = \begin{cases} 0 & \text{if } x \leq 0, \\ 2x & \text{if } x > 0 \end{cases}$

(i) Sketch the function $f(x)$, from $x = -2$ to $x = 2$ 2

(ii) Evaluate $\int_{-2}^2 f(x) dx$ 1

b) The function $f(x)$ is defined by the rule $f(x) = 9x(x-2)^2$ in the domain $-1 \leq x \leq 3$.

(i) find the x and y intercepts 2

(ii) find the stationary points and determine their nature. 3

(iii) find the values of the end-points 2

(iv) draw a sketch of the graph of $y = f(x)$, showing clearly the turning points, the intercepts and the end-points. 2

Question 8**Marks**

a) Evaluate the following integrals:

6

(i) $\int_1^2 \frac{1}{x^3} dx$

(ii) $\int_1^4 e^{3x} dx$

(iii) $\int_0^{\frac{\pi}{8}} \sec^2 2x \, dx$

b) Find $\int \frac{x}{x^2 + 4} dx$

2

c) The population P of a town is growing at a rate proportional to the town's current population. The population at time t years is given by $P = Ae^{kt}$, where A and k are constants.

The population 20 years ago was 100 000 people and today the population of the town is 150 000 people.

(i) Find the value of A

1

(ii) Find the value of k

1

(iii) Find the population that will be present 20 years from now.

2

Question 9

a) Find $\frac{dy}{dx}$ given that $y = \log_e \left(\frac{2x+1}{3x-7} \right)$

2

b) A particle moves in a straight line so that its velocity, v metres per second, at time t is given by $v = 3 - \frac{2}{1+t}$.

The particle is initially 1 metre to the right of the origin.

(i) Find an expression for the position x , of the particle at time t .

2

(ii) Explain why the velocity of the particle is never 3 metres per second

1

(iii) Find the acceleration of the particle when $t = 2$ seconds.

2

c) (i) Show that $(\operatorname{cosec}^2 A - 1)\sin^2 A = \cos^2 A$.

2

(ii) Hence, or otherwise solve $(\operatorname{cosec}^2 A - 1)\sin^2 A = \frac{3}{4}$ for $-\pi \leq A \leq \pi$

3

Question 10

Marks

An open cylindrical container is to hold 16m^3 of grain.

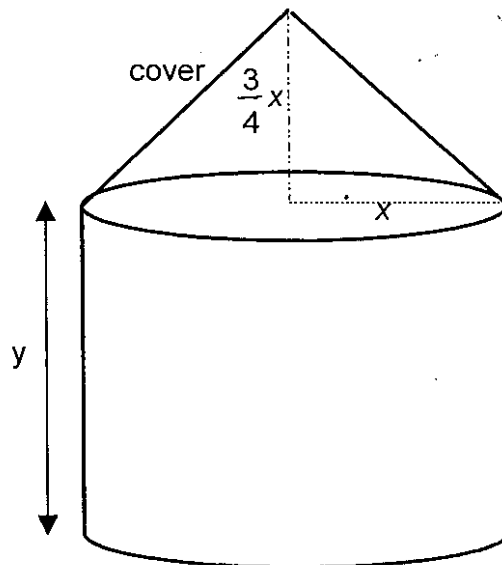
- a) If the radius of the container is x metres and it's height is y metres, show that 2

$$y = \frac{16}{\pi x^2}.$$

Hence show that the surface area, in square metres, of the container (sides and base) is 3

$$\pi x^2 + \frac{32}{x}.$$

- b) A conical cover of height $\frac{3}{4}x$ metres is placed on top to form a silo. Given that the 3
surface area of an open cone of radius r , and slant height s is πrs , show that the surface
area in square metres, of this cover is $\frac{5\pi x^2}{4}$.



- c) The cost per unit area of the cover is 50% more than the cost per unit area of the sides and base. If k dollars per square metre is the cost per unit area of the sides and base, show that the total cost C in dollars of the silo (cover, sides and base) is given by

$$C = k\left(\frac{23}{8}\pi x^2 + \frac{32}{x}\right) \quad 2$$

- d) Find the value of x which minimises the total cost. 2

QUESTION 1

a) $1.488437459 = 1.49$ ✓

b) $x^3 = 4x^2$

$$x^3 - 4x^2 = 0$$

$$x^2(x-4) = 0$$

$$\therefore x = 0, x = 4$$

c) $\frac{1}{3} e^{-x} \Rightarrow -\frac{1}{3} e^{-x} = \frac{-1}{3e^x}$ ✓

d) $\frac{1}{m^2 - 4m + 3} - \frac{1}{m^2 - 1}$
 $\frac{1}{(m-1)(m-3)} - \frac{1}{(m+1)(m-1)}$

$$\frac{m+1 - m+3}{(m-1)(m-3)(m+1)}$$

$$= \frac{4}{(m-1)(m+1)(m+3)}$$
 ✓

e) $\frac{1}{y} - 2y = 1 \Rightarrow 1 - 2y^2 = y$
 $0 = 2y^2 + y - 1$
 $(2y-1)(y+1) = 0$

$$y = \frac{1}{2}, y = -1$$

$$x = 2, x = -1$$
 ✓

f) $\frac{1}{2-\sqrt{3}} \times \frac{2+\sqrt{3}}{2+\sqrt{3}} = \frac{2+\sqrt{3}}{4-3} = 2+\sqrt{3}$

$$\therefore a = 2, b = 1$$
 ✓

$$\begin{aligned}
 & \frac{3x}{\sin x} \Rightarrow \frac{3x^2 e^x + x^3 e^x}{\sin^2 x} = \frac{21(3x+4)^6}{x^2 e^{2x} (3+x)} \\
 & \Rightarrow \frac{3 \sin x - 3x \cos x}{\sin^2 x}
 \end{aligned}$$

$$\begin{aligned}
 & \tan \frac{\pi}{3} + \operatorname{cosec} \frac{\pi}{4} \\
 & = \tan 60 + \frac{\sqrt{2}}{1} \\
 & = \sqrt{3} + \sqrt{2}
 \end{aligned}$$

$$\begin{aligned}
 c) \quad & x^2 - (k+2)x + 4 = 0 \\
 & \Delta \geq 0 \\
 & (k+2)^2 - 4(1)(4) \geq 0 \\
 & (k+2)^2 \geq 16 \\
 & k+2 \geq 4
 \end{aligned}$$

$$\begin{aligned}
 & \therefore k \geq 4, \quad k \leq -4 \\
 & -6 \geq k \geq 2
 \end{aligned}$$

$$\begin{aligned}
 d) \quad & \frac{\sin R}{18.6} = \frac{\sin 37}{12.5} \\
 & \sin R = \frac{18.6 \times \sin 37}{12.5} \\
 & R = 116^\circ 26'
 \end{aligned}$$

QUESTION #3

$$\begin{aligned}(i) \quad AB &= \sqrt{(5-1)^2 + (2-4)^2} \\&= \sqrt{36+4} \\&= \sqrt{40} \\&= 2\sqrt{10}\end{aligned}$$

$$(ii) \quad M = \left(\frac{5+1}{2}, \frac{4+2}{2} \right) = (2, 3)$$

$$(iii) \quad m = \frac{2}{-6} = -\frac{1}{3}$$

$$(iv) \quad Y-4 = -\frac{1}{3}(X+1)$$

$$3Y-12 = -X-1$$

$$3Y = -X+11$$

$$\Rightarrow X+3Y-11=0$$

$$(v) \quad \text{Gradient of } L = +3 \quad \therefore M_{AB} \times M_L = 3 \times -\frac{1}{3} = -1$$

$$\text{and } (2, 3) \text{ lies on } L \quad [(3(2) - 3 - 3 = 0)]$$

$$(vi) \quad X\text{-intercept, } Y=0$$

$$\begin{aligned}3(0) - Y - 3 &= 0 \\-3 &= 0\end{aligned}$$

$$3X - 0 - 3 = 0$$

$$3X = 3$$

$$X = 1$$

$$\therefore C \text{ is } (1, 0)$$

$$(v) \quad \text{Equation of circle:}$$

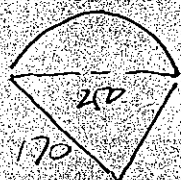
$$(X-2)^2 + (Y-3)^2 = 10$$

$$b) \quad (i) \quad \alpha + \beta = 6$$

$$(ii) \quad \alpha\beta = 10$$

$$(iii) \quad (\alpha+1)(\beta+1) = \alpha\beta + \alpha + \beta + 1 = 6 + 10 + 1 = 17$$

QUESTION 4



$$(i) \cos \theta = \frac{170^2 + 170^2 - 280^2}{2(170)(170)} = -0.08134878$$

$$\theta = 95^\circ$$

$$(ii) \text{ Arc length} = \frac{95}{360} \times 2 \times \pi \times 170 = 282 \text{ km.}$$

$\therefore$ The distance was shortened by $282 - 280 = \underline{32 \text{ km.}}$

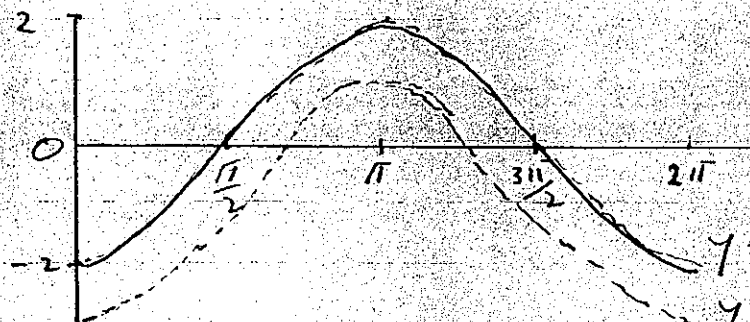
$$b) 16y = x^2 \quad 49 = 16$$

$$a = 4$$

$$(i) (0, 4)$$

$$(ii) y = -4$$

$$c) (i) y = -2 \cos x$$



$$y = -2 \cos x$$

$$y = -2 \cos x - 1$$

QUESTION 5

a) (i) $y = x \sin x \quad \frac{dy}{dx} = \sin x + x \cos x$

At $\frac{\pi}{2}$, $\frac{dy}{dx} = 1 + \frac{\pi}{2} \cdot 0 = 1$

Equation of tangent $y - \frac{\pi}{2} = 1(x - \frac{\pi}{2})$

(ii) $y = x$ ✓
Equation of Normal:

$$y - \frac{\pi}{2} = -1(x - \frac{\pi}{2})$$

$$y = -x + \pi \quad \checkmark$$

b) $V = \pi \int_0^{\pi/4} (\sec x)^2 dx$

$$= \pi \left[\tan x \right]_0^{\pi/4}$$

$$= \pi [\tan \frac{\pi}{4} - \tan 0]$$

$$= \pi \text{ unit}^3 \quad \checkmark$$

c) $A = \frac{1}{3} [0 + 3 + 2(5) + 4(2+4)]$

$$= \frac{1}{3} [3 + 10 + 24]$$

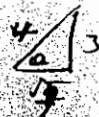
$$= \frac{37}{3}$$

$$= 12\frac{1}{3}$$

0	1	2	3	4
0	2	5	4	3
0	2	5	4	3

QUESTION 6

a) (i) $\cos \theta = \frac{\sqrt{7}}{4}$ ✓



(ii) $\cos \theta + \tan \theta = \frac{\sqrt{7}}{4} + \frac{3}{\sqrt{7}} = \frac{\sqrt{7}}{4} + \frac{3\sqrt{7}}{7}$
 $= \frac{7\sqrt{7} + 12\sqrt{7}}{28}$
 $= \frac{19\sqrt{7}}{28}$ ✓

b) $\sin 2\theta = 1$
 $2\theta = \frac{\pi}{2}, \frac{5\pi}{2}$
 $\theta = \frac{\pi}{4}, \frac{5\pi}{4}$ ✓

c) $x^2 - 4x > 0$
 $x(x-4) > 0$
 $x < 0, 4$



$0 \leq x \leq 4$

d) $y = m(x+1)$ $y = 2x^2$ $\therefore 2x^2 = mx + m$
 $= mx + m$

$2x^2 - mx - m = 0$

$\therefore$ No intersection, $b^2 - 4ac < 0$

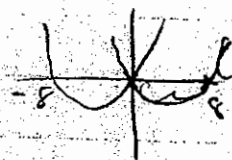
$m^2 - 4(2)(-m) < 0$

$m^2 + 8m < 0$

$m(m+8) < 0$

$m = 0, m = -8$ $-8 < m < 0$ ✓

$-8 < m < 0$



e) $e^x - 9e^{-x} = 0$

$e^x - \frac{9}{e^x} = 0$

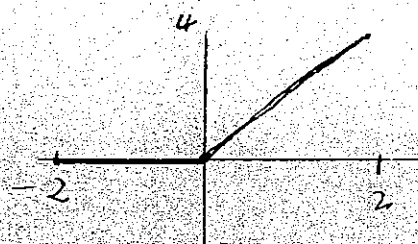
$\frac{e^{2x} - 9}{e^x} = 0$

$e^{2x} = 9$

$x = \frac{\ln 9}{2}$

QUESTION 7

a) (i)



$$(ii) \int_{-2}^2 f(x) dx = \frac{1}{2} \times 2 \times 4 = 4 \text{ unit}^2$$



(b) $f(x) = 9x(x-2)^2$

(i) y-int, $x=0 \quad \therefore 9(0)(0-2)^2 = 0$

x-int, $y=0 \quad \therefore 9x(x-2)^2 = 0 \quad \therefore x=0, 2$ ✓

$$\begin{aligned} (ii) f'(x) &= 9(x-2)^2 + 9x[2(x-2)] \\ &= 9(x-2)^2 + 9x(2x-4) \\ &= 9(x^2 - 4x + 4) + 18x^2 - 36x \\ &= 9x^2 - 36x + 36 + 18x^2 - 36x \\ &= 27x^2 - 72x + 36 \end{aligned} \quad \left| \begin{array}{l} 9x^2 - 4x + 4 \\ 9x^3 - 36x^2 + 36x \\ 27x^2 - 72x + 36 \end{array} \right.$$

$f'(x) = 0 \quad \therefore 27x^2 - 72x + 36 = 0 \quad \div 9$

$$3x^2 - 8x + 4 = 0$$

$$(3x-2)(x-2) = 0$$

$$x = \frac{2}{3}, x = 2$$

$$\begin{aligned} f\left(\frac{2}{3}\right) &= 9\left(\frac{2}{3}\right)\left(\frac{2}{3}-2\right)^2 \\ &= \frac{18}{3} \cdot \frac{32}{9} = \left(\frac{272}{3}\right) \quad \left| \quad 9(2)(2-2)^2 = 0 \quad (20) \right. \end{aligned}$$

Nature: $f''(x) = 54x - 72$

$f''\left(\frac{2}{3}\right) = \frac{18}{3}\left(\frac{2}{3}\right) - 72$ ✗

(MA)

$f''(2) = 54(2) - 72$ ✓

MI

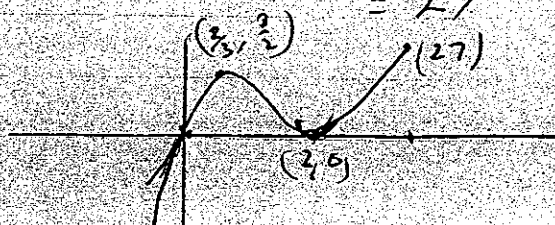
$f(-1) = 9(-1)(-1-2)^2$

$= -9(9)$

$= -81$ ✓

$f(3) = 9(3)(3-2)^2$

$= 27$ ✓



QUESTION 8

$$9) (i) \int_1^2 x^3 dx = \left[\frac{-1}{2x^2} \right]_{-1}^2 = \left[\frac{-1}{8} + \frac{1}{2} \right] = \frac{3}{8} \quad \checkmark$$

$$(ii) \int_1^4 e^{3x} dx = \left[\frac{1}{3} e^{3x} \right]_1^4 = \left[\frac{1}{3} e^{12} - \frac{1}{3} e^3 \right] \\ = \frac{1}{3} e^3 (e^9 - 1) \quad \checkmark$$

$$(iii) \int_0^{\pi/8} \sec^2 2x dx = \left[\frac{1}{2} \tan 2x \right]_0^{\pi/8} = \frac{1}{2} \quad \checkmark$$

$$b) \int \frac{x}{x^2+4} dx = \frac{1}{2} \ln(x^2+4) + C \quad \checkmark$$

$$c) i) A = 100\,000 \quad \checkmark$$

$$(ii) 150\,000 = 100\,000 e^{k(20)} \\ 1.5 = e^{20k} \\ \frac{\ln 1.5}{20} = k \quad \checkmark$$

$$(iii) P = 100\,000 e^{\frac{\ln 1.5}{20} \times 40} \\ = 100\,000 e^{2 \ln 1.5} \\ = \underline{225\,000} \quad \checkmark$$

QUESTION 9

$$a) \quad y = \log \frac{2x+1}{3x-7} = \log(2x+1) - \log 3x-7$$

$$y' = \frac{2}{2x+1} - \frac{3}{3x-7}$$

$$b) \quad v = 3 - \frac{2}{1+t}$$

$$x = 3t - 2 \ln(1+t) + C$$

$$x = 0 \text{ when } t = 0$$

$$\therefore 1 = 3(0) - 2(\ln 1) + C$$

$$1 = C$$

$$(1) \quad \therefore x = 3t - 2 \ln(1+t) + 1$$

(ii) Since $\frac{2}{1+t}$ can never be 0, v will never be 3.

$$(iii) \quad v = 3 - 2(1+t)^{-1}$$

$$a = 2(1+t)^{-2} = \frac{2}{(1+t)^2}$$

$$a(2) = \frac{2}{(1+2)^2} = \frac{2}{9} \text{ m/sec}^2$$

$$c) \quad (i) \quad \text{L.H.S.} \quad (\operatorname{cosec}^2 A - 1) \sin^2 A$$

$$= \left(\frac{1}{\sin^2 A} - 1 \right) \cdot \sin^2 A$$

$$= \frac{1 - \sin^2 A}{\sin^2 A} \cdot \sin^2 A = 1 - \sin^2 A = \cos^2 A$$

$$(ii) \quad \cos^2 A = \frac{3}{4}$$

$$A = \cos^{-1} \pm \frac{\sqrt{3}}{2}$$

$$A = \frac{\pi}{6}, -\frac{\pi}{6}$$

QUESTION 10

$$\begin{aligned} a) \quad V &= \pi r^2 h \\ 16 &= \pi x^2 y \end{aligned}$$

$$\frac{16}{\pi x^2}$$

$$= y$$

$$S.A = \pi r^2 + 2\pi r h$$

$$\begin{aligned} &= \pi x^2 + 2 \pi x \cdot \frac{16}{\pi x^2} \\ &= \pi x^2 + \frac{32}{x} \end{aligned}$$

$$\begin{aligned} b) \quad s &= \left(\frac{3}{4}x\right)^2 + x^2 \\ &= \frac{9}{16}x^2 + x^2 \\ &= \frac{25x^2}{16} \end{aligned}$$

$$s = \frac{5x}{4}$$

$$\begin{aligned} S.A &= \pi r s \\ &= \pi \times x \cdot \frac{5x}{4} \\ &= \frac{5\pi x^2}{4} \end{aligned}$$

$$\begin{aligned} c) \quad C &= K \times \left(\pi x^2 + \frac{32}{x}\right) + 1.5K \left(\frac{5\pi x^2}{4}\right) \approx \frac{3K}{2} \left(\frac{5\pi x^2}{4}\right) \\ &= K\pi x^2 + \frac{32K}{x} + \frac{15K\pi x^2}{8} \\ &= \frac{23K\pi x^2}{8} + \frac{32K}{x} \\ &= K \left(\frac{23\pi x^2}{8} + \frac{32}{x} \right) \end{aligned}$$

$$d) \quad \frac{dC}{dx} = \frac{46K\pi x}{8} - \frac{32K}{x^2}$$

$$0 = \frac{46K\pi x^3}{8x^2} - 256K$$

$$46\pi x^3 = 256$$

$$\text{At Min } \frac{dC}{dx} = 0$$

$$x = \sqrt[3]{\frac{256}{46\pi}}$$

$$= 1.210$$